



Validation of Garmin and Polar Devices for Continuous Heart Rate Monitoring During Common Training Movements in Tactical Populations

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2 During Common Training Movements in Tactical Populations

3 Abstract

4 Heart rate samples (n=4500-8000) from wearables were compared to electrocardiography during a steady-
5 state ruck (Ruck-S), maximal effort ruck (Ruck-M), submaximal cycle (Cycle), and Tabata Circuit. One
6 device was worn at each location (wrist: Polar Grit-X, Garmin Fenix 6; chest-straps: Polar H10, Garmin
7 HRM-Pro; armband: Polar Verity). Comparisons were made via percent error (MAPE) $\leq 5\%$, Concordance
8 Correlation Coefficient (CCC), and ordinary least products regressions (OLP). H10 demonstrated strong
9 agreement for all movements (MAPE= 1.28-3.40%, CCC= 0.93-0.99). During Ruck-S, Ruck-M, and Cycle,
10 HRM-Pro (MAPE= 1.96-3.73%, CCC= 0.95-0.99) and Verity (MAPE= 1.84-5.36%, CCC= 0.98-0.99)
11 demonstrated strong agreement. Fenix-6 demonstrated low MAPE (4.23-5.44%) and moderate to strong
12 CCC (0.76-0.96) for Ruck-S, Ruck-M, and Cycle, while Grit-X had poor agreement (MAPE= 8.49-16.45%;
13 CCC= 0.24-0.78). Tabata Circuit had worst disagreement for all devices. Overall, chest straps and armbands
14 demonstrated the strongest agreement, and should be worn when precise heart rate training is necessary.

15 Key Words: wearables; validity; accuracy; tactical; ECG

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17 INTRODUCTION

18 When properly implemented, heart rate data may help enhance high performing populations
19 (e.g., military) by reducing overtraining, increasing preparedness, and mitigating risk of attrition
20 and injury through adjustments of physical training and recovery modalities (Murdock & Hagen,
21 2018). Summary heart rate data can quantify session, daily, weekly, and total physiological
22 training stress endured during military training and occupational tasks (Jurvelin et al., 2020).
23 However, real-time monitoring of continuous heart rate data, often noted as a percentage of relative
24 maximal heart rates, identifies individualized physiological states (Dijksma et al., 2021; Maupin
25 et al., 2019) and metabolic demands of events such as rucksack marching/ rucking (Simpson et al.,
26 2010). Biofeedback via physiological technologies allows individuals to effectively use heart rate
27 goals to achieve their intended training adaptations or to quantify their likelihood completing
28 endurance activities (e.g., rucking) at their current pace (Oviatt, 2017). The heart rate feedback
29 may be directly monitored by the individual to maintain prescribed exercise intensities or, in some
30 instances, can be displayed to central hub (e.g., Polar Team Pro) using a for a group-based
31 evaluation by the practitioner. Regardless of the deployment method, successful continuous
32 monitoring is contingent on wearable sensors' ability to measure heart rate validly and feasibly.

33 The use of wearable sensors (i.e., smart watches and bands) has exponentially grown in recent
34 years, becoming the top fitness industry trend in 2022 with a 100 billion dollar valuation
35 (Thompson, 2022). Some commercially available physiological wearable sensors are medical or
36 research-oriented (i.e., electrocardiography, ECG) and considered criterion devices, but are often
37 more difficult to utilize and wear in a non-clinical setting (Lee et al., 2014). Consumer-grade
38 wearables provide increased usability in the common form of wrist worn devices. However,
39 regardless of their potential for producing low-fidelity data (Lee et al., 2014), these devices are

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being implemented to inform decisions in training and free-living conditions for soldiers, athletes, general, and clinical populations (Li et al., 2016; Murdock & Hagen, 2018; Seshadri et al., 2019). This is a major concern since these devices are used to characterize physiological statuses, which may be incorrect depending on device validity in various tasks and environments.

Typically, chest strap devices result in superior validity and less sensitivity to variations in activity intensity (Cosoli et al., 2020; Etiwy et al., 2019; Gillinov et al., 2017; Pasadyn et al., 2019), which may stem from their ability to stay in position with less motion artifact throughout various movements, the use of electrocardiography compared to photoplethysmography (PPG) signals, and the proximity of the sensors to the heart (Murdock & Hagen, 2018). Some have found wrist-worn PPG based devices to be accurate during sedentary or low-intensity tasks (e.g., walking, cycling), but the validity often decreases as tasks increase in intensity or involve ballistic movements (Boudreaux et al., 2018; Chow & Yang, 2020; Düking et al., 2020; Hajj-Boutros et al., 2021; Murdock & Hagen, 2018; Pasadyn et al., 2019). Errors in PPG devices may coincide with changes in blood flow around the wrist area from wrist flexion or extension, motion artifacts from excessive or rapid arm movements, or tightly grasping an object (Bent et al., 2020; Butler et al., 2016; Horton et al., 2017; Murdock & Hagen, 2018). Similarly, in resistance exercise, wrist-worn devices seem to possess less validity as exercise intensity increases (Boudreaux et al., 2018; Jo et al., 2016; Müller et al., 2019). However, since technology manufacturers continuously release new models and discontinue existing models, research concerning the validity of wearable devices requires constant exploration across various form-fitting devices and conditions (Fuller et al., 2020).

When doing so, second-by-second heart rate comparisons should be made to reflect real-time monitoring, which to date, few studies have accomplished (Chow & Yang, 2020; Gillinov et al.,

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2017; Hettiarachchi et al., 2019; Navalta et al., 2020). These studies should also be conducted for more realistic training durations (i.e., ~15 minutes compared to ~2-5 minutes) to provide enough time to capture any possible drops in validity throughout real world applications. The purpose of the current study was to determine the validity of continuous heart rate data from wearables, currently used by high performing populations, in comparison to ECG during tactically relevant movements such as 1) submaximal cycling with increasing intensities while maintaining constant cadence (purpose to capture data in all heart rate zones), 2) circuit-style (i.e., Tabata) upper and lower body training with fluctuations in intensity (purpose to reflect common high intensity interval training common in tactical populations), 3) submaximal steady-state loaded ruck marching (purpose to reflect very common tactical exertion), 4) incremental to maximal effort ruck marching (purpose to reflect a maximal intensity common tactical movement).

MATERIALS AND METHODS

Participants

Seven healthy men (age= 27 ± 8 years; height= 173.0 ± 7.0 cm; body mass= 72.7 ± 6.8 kg; body fat= $16.0 \pm 4.5\%$; body mass index= 24.5 ± 1.7 kg $\times$ m⁻²) and one healthy woman (age= 29 years; height= 160.0 cm; body mass= 62.2 kg; body fat= 25.2%; body mass index= 24.3 kg $\times$ m⁻²) participants volunteered for this study. Each participant was recreationally active (regular participation in sport, cardio exercise, or resistance training) for at least 6 consecutive months and without a musculoskeletal injury within the last 6 months leading up to testing. Key exclusion criteria included having tattoos at the site of measurement for sensor placement, pregnancy or actively trying to become pregnant, or having an infectious skin disease. Participants identified themselves as Caucasian and had a skin tone at the site of optical sensor placement (e.g., top of the wrist) of 1-2 on the Fitzpatrick Phototyping Scale (Fitzpatrick, 1988). Participants wore

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exercise attire similarly resembling physical training uniforms of tactical populations, such as athletic shorts and a t-shirt or tank top. Prior to initiating data collection, participants were informed of the risks and benefits of the study and signed the institutionally approved informed consent document. All procedures contained herein were approved by the Institutional Review Board of West Virginia University (#2103275452) for human subject's research and were compliant with the Declaration of Helsinki guidelines.

Procedures

The criterion validity of wearable sensors for measuring heart rate data was assessed during a series of tactically relevant aerobic exercises. During testing, participants donned one wrist-worn device on each wrist (Polar Grit, Garmin Fenix 6) and one chest-strap (Polar H10, Garmin HRM-Pro) or armband (Polar Verity) in addition to the reference multi-lead ECG device (Bittium Faros 180™). One device per location was allowed to prevent for any potential signal interference. Thus, participants completed testing on multiple days to allow each participant to don every device appropriately (e.g., Polar H10 chest strap for one session and Garmin HRM-Pro for another session). Wearable devices were worn in random order but kept the same across testing sessions (R, Rucking; C, Cycling and Circuit Training), which were all separated by at least 48 hours. The devices were all placed on the participants and verified for appropriate location by the same investigators.

Participants completed the protocol testing sessions in counterbalanced order (i.e., 4 participants began with R protocols, 4 began with C protocols). During two R sessions, participants completed a steady-state rucking protocol (RUCK-S) followed by a maximal incremental graded rucking protocol to volitional exhaustion (RUCK-M). During two C sessions, participants completed a submaximal cycle ergometer test (CYCLE) followed by Tabata-style circuit training

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(Tabata-Circuit). Each session began with a brief 3-minute low-intensity stage to serve as the warm-up to RUCK-S and CYCLE. There was a 10-minute rest provided between protocols within the R and C sessions. The layout of these protocols is demonstrated in Figure 1. All participants were familiar with wearing weighted vests during walking or running tasks and had experience with Tabata-style circuits. Prior to the C sessions beginning, participants were shown the exercises for the Tabata-Circuit and allowed practice to ensure they were familiar with each exercise.

Insert Figure 1 about here

Treadmill Rucking Protocols

Participants completed RUCK-S followed by RUCK-M on a treadmill (4Front, Woodway, Waukesha, WI, USA). RUCK-S began with a 4.8-kph pace at 0% incline for 3 minutes, followed by a 6.4-kph pace at 0% incline for 12 minutes for a total of 15 minutes. Participants then rested for 10 minutes before completing RUCK-M, which progressed across 3-minute stages (Figure 1), as described previously (Swearingen et al., 2018). RUCK-M was completed to maximal volitional exhaustion, or the point in which the participant felt and/or expressed an inability to continue exercise. Participants were instructed to maintain a traditional rucking gait, such that if participants began to jog/ run testing ceased. Participants wore a ~15 kg weighted vest as external load to replicate tactical equipment. The vest was snugly positioned on their torso with equal weight distribution to the front and back of the kit. The adjustment location of the vest (strap to tighten fit) was below the chest strap monitors and ECG leads placing limited direct pressure and friction / contact artefact on device signals.

Cycle Ergometer and Tabata Circuit Protocol

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Participants completed a graded submaximal cycle ergometer protocol (CYCLE) followed by a Tabata-style circuit workout (Tabata-Circuit) with a 2:1 work-to-rest ratio in 30-second intervals. Cycle ergometer (894E, Monark, Varberg, Sweden) testing followed the YMCA protocol (Garatachea et al., 2007), which consisted of a 3-minute warm-up with 0 resistance followed by 4 stages of 3-4 minutes at 50 rpm. Stage 1 had 3 minutes of cycling at 0.5 kg resistance with an added fourth minute if the difference in heart rate between minutes 2 and 3 exceeded 5 bpm. The final heart rate during Stage 1 determined the workload for Stages 2, 3, and 4 (Figure 1). Stages 2-4 also consisted of 3 minutes with an additional 4th minute if there was a difference >5 bpm between the heart rate at minutes 2 and 3. Exercise was terminated once the participant reached 85% of their age-predicted maximal heart rate ($220 - \text{age}$). It should be noted that three participants did not reach 85% of their maximal heart rate, and resultantly, their testing ceased at the 3rd or 4th minute of the 4th stage.

The Tabata-Circuit was programmed as a simple replication of potential functional exercises that may be performed in various settings (box gym, traditional gym, at home gym) and include rapid arm motions or grasping tasks. Participants completed two groups of four movements, lasting ~18 minutes in total. The first group of movements included pull-ups, kettlebell swings (woman, 16 kg; man, 24 kg), push-ups, and farmer's carry (24 kg), while the second included burpees, sit-ups, box jumps (60 cm), and air squats. Participants completed the first group of movements followed by a 1-minute passive rest then completed the second group of movements followed by another 1-minute rest for three complete rounds. Each movement was performed for 20 seconds, followed by 10 seconds of rest. All exercises were executed with maximal effort for the full 20-second bout by attempting to perform as many repetitions as possible. During recovery between exercises participants actively rested by walking to and preparing for the subsequent exercise.

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153 Between groups of movements, the participants were permitted to drink water but asked to
154 otherwise limit movement during the passive rest period.

155 Reference Holter Electrocardiography Device

156 An ambulatory ECG device (Bittium Faros 180, Bittium Inc., Oulu, Finland) was used as the
157 criterion reference device (Bent et al., 2020). The 3 ECG monitoring electrodes (Red Dot 2570,
158 3M™, Center St. Paul, MN, USA) were placed under the right and left clavicle bones as well as
159 in a left intercostal space (between two ribs on the left side of the body) for the duration of each
160 testing session. Skin was prepped at the placement sites for each electrode by shaving any hair and
161 cleansing with slight abrasion with alcohol wipes to minimize noise artifacts and improve signal
162 quality. If the ECG lead wires appeared loose, adhesive tape was used to secure them to the
163 participant's body to minimize motion artifacts from electrode movement. The locations were
164 marked with permanent marker, and participants were asked to maintain the marking if it were to
165 fade, to keep placement consistent across sessions within the same participants.

166 Exports (.edf format) from the reference ECG device were exported from the Bittium Faros
167 Software and imported into Kubios HRV Premium Software (v3.4.3, Kubios, Kuopio, Finland)
168 for analysis (Tarvainen et al., 2014, 2021). The R Peaks were automatically detected, and the
169 intervals between successive R Peaks (R-R Intervals) were calculated through the intrinsic QRS
170 detection algorithm of the Kubios HRV premium software. Identified error corrections were <5%
171 for all trials. Noise segments were automatically detected using the Kubios HRV Premium
172 software (based on R-R intervals) with noise detection level set at medium to clean data where
173 poor signal quality may have occurred. Further, the automatic artifact correction algorithms were
174 used, which have previously been validated (Lipponen & Tarvainen, 2019). Additionally, the ECG
175 waveforms were visually inspected to verify R wave locations and motion artifacts. The

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measurement rate was set to 1000 Hz, time series interpolation rate was set to 4 Hz, and RR detrending method was set to Smooth n priors at 500 with a 0.035 Hz cutoff frequency. The beat-to-beat heart rate from the criterion device were derived to match the same sampling frequency of the consumer devices as second-to-second heart rate.

Study Instruments Being Tested

The devices analyzed for the current study included: two optical-based wrist-worn devices [Fenix 6 (Garmin, Olathe, Kansas, United States), Grit-X (Polar, Kempele, Finland)], two ECG-based chest strap devices [HRM-Pro (Garmin, Olathe, Kansas, United States), H10 (Polar, Kempele, Finland)], and one optical-based armband device [Verity Sense (Polar, Kempele, Finland)]. Each device was worn according to manufacturer guidelines. The devices were all synced via Bluetooth to a commercially developed smartphone application respective to each device (i.e., Polar Beat, Polar Flow, Garmin Connect). Data were then exported from the applications to Microsoft Excel files, or as TCX files from Garmin Connect and transferred to Excel, for analysis.

The wrist and armband devices were randomly ordered for each participant in terms of which sessions they would be worn and on which wrist/arm they would be worn. The randomization was set to be counterbalanced such that the wrist and arm worn devices had probability of 0.5 for being assigned to the left or right arm for each participant. The Verity armband device was worn on the brachial midline such that the sensor placed in the armband holder was pressed firmly against the epidermis. The wrist-worn devices were worn according to manufacturer guidelines with all their sensors making skin contact with the ventral side of the carpus/wrist and placed proximally to the radial and ulnar notches. Chest strap devices, H10 and HRM-Pro, were fastened with the center just inferior to the xiphoid process and the electrode areas moistened prior to donning.

Statistical Analyses

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199 All second-by-second heart rate data across devices were aligned using the recorded start and
200 end times for each protocol. Sample sizes for each device and protocol varied slightly due to some
201 trials being longer across protocols and signal connectivity of devices. All wrist-, arm-, and chest
202 strap-based devices were compared to the reference ECG using Ordinary Least Products (OLP)
203 regressions (Model II linear regression; “sma” function within the “smatr” package; Ludbrook,
204 2010). Systematic bias was present when the 95% confidence interval of the intercept did not
205 include “0”, while proportional bias was present when the 95% confidence interval of the slope
206 did not include “1.0”. Additionally, agreement was tested using Lin’s Concordance Coefficients
207 (CCC) with a minimal acceptable cutoff of 0.80 based on previous heart rate validation studies
208 (Gillinov et al., 2017; Pasadyn et al., 2019). Further cutoffs included 0.90 to 0.95: moderate, 0.95
209 to 0.99: substantial, > 0.99: almost perfect (Lin, 1989; McBride, 2005). Agreement was also tested
210 visually via Bland-Altman plots using the 95% limits of agreement technique (mean bias $\pm$ [1.96
211 $\times$ SD of Differences]) with respective confidence intervals. Lastly, mean absolute percent error
212 (MAPE), as $[(\text{ECG} - \text{Device})/\text{ECG}] \times 100$ was calculated from each second of data throughout
213 a protocol. A MAPE of >5% was considered unacceptable and the MAPE results were compared
214 across devices using repeated measures analysis of variance followed by pairwise comparisons
215 with Bonferroni corrections. All statistical procedures were conducted using R, version 4.0.1 (R
216 Core Team, Vienna, Austria; <https://www.R-project.org>) with alpha level set to < 0.05.

217 RESULTS**218 Steady-state Treadmill Rucking (RUCK-S)**

219 There was systematic bias for all devices except H10 and Fenix-6 (Table 1). According to OLP
220 regression slopes and Bland-Altman plots, proportional bias existed for all devices except H10
221 (Figure 2). Poor agreement was noted for Fenix-6 and Grit-X according to low associations and

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unacceptable CCC (Table 1). HRM-Pro demonstrated moderate agreement, Verity demonstrated substantial agreement, and H10 demonstrated nearly perfect agreement (Table 1). The MAPE of all devices were considered acceptable, while wrist-worn devices had wide error limits (Table 1). The H10 and Verity were not significantly different than each other and had statistically lower MAPE ($p<0.001$) than all other devices, while the Grit-X had the highest MAPE of all devices ($p<0.001$).

Insert Table 1 and Figure 2 about here

Incremental Maximal Treadmill Rucking (RUCK-M)

There was systematic bias for all devices, but MAPE was low for H10, HRM-Pro, Verity, and Fenix-6 (Table 2). According to OLP regression slopes and Bland-Altman plots, proportional bias existed for all devices except H10 (Figure 3). Poor agreement was noted for Grit-X according to low associations and unacceptable CCC (Table 2). H10, HRM-Pro, and Verity demonstrated nearly perfect agreement and high associations, while Fenix-6 agreement was substantial (Table 2). The H10 had statistically lower MAPE ($p<0.001$) than all other devices, while the Grit-X had the highest MAPE of all devices ($p<0.001$). The Garmin HRM-Pro and Polar Verity MAPE were not significantly different (Table 2).

Insert Table 2 and Figure 3 about here

Submaximal Cycling (CYCLE)

The H10, HRM-Pro, Verity, and Fenix-6 all demonstrated strong associations and substantial agreement according to Lin's Concordance Coefficients (Table 3). The H10 revealed small systemic underestimations, while the Fenix-6 demonstrated small systemic overestimations (See MAPE and OLP Intercept in Table 3, as well as the Bland-Altman plot in Figure 4). Further, the

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systematic bias of the H10 did not change as heart rate increased, according to the lack of proportional bias noted in the OLP regressions and Bland-Altman plots (Figure 4), whereas the small errors of the HRM-Pro and Verity were not systematic and were dependent on heart rate magnitudes (Table 3). Although the bias was not systematic for the Grit-X, this device demonstrated unacceptable agreement, significantly higher MAPE than all other devices, and proportionate bias (Table 3, Figure 4). The Polar H10 and Garmin HRM-Pro MAPE were lowest and were not significantly different than each other (Table 3).

Insert Table 3 and Figure 4 about here

Tabata-Circuit

There was systematic bias for all devices except H10 (Table 4), while proportionate bias existed for all devices (Figure 5). Poor agreement was noted for Fenix-6 and Grit-X according to low associations and unacceptable CCC (Table 4). H10 and HRM-Pro demonstrated moderate agreement, while Verity demonstrated poor agreement with large error limits (Table 4). The MAPE of all devices was considered acceptable, except Fenix 6 and Grit-X which had statistically higher MAPE than all other devices ($p < 0.001$; Table 4). The Polar H10 and Garmin HRM-Pro MAPE were lowest and were not significantly different than each other (Table 4).

Insert Table 4 and Figure 5 about here

DISCUSSION

This study contributes to the ongoing requirement for feasibility and validity testing of wearable sensors, in controlled yet applicable environments to the target user population, which in this study was the tactical community (military, first responder). Further, this study investigated several devices from Garmin and Polar manufacturers that are commonly used in athletic and

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tactical settings (Gilgen-Ammann et al., 2019; Li et al., 2016), where heart rate zone-based training may be deployed, by using unique structures and longer durations than previous research validating wearable devices. Across all exercises, the H10 demonstrated the strongest agreement with the reference ECG device, followed by the HRM-Pro with slight underestimations (mean bias = -1.53 to -2.63 bpm) and the Verity with slight overestimations (mean bias = 1.69 to 1.77 bpm) during the rucking tasks which were statistically but not practically significant. Comparing device groups, wrist-worn devices demonstrated the weakest agreement overall, but Fenix-6 demonstrated superior agreement with the ECG compared to the Grit-X for all exercises. Tabata-Circuit resulted in the weakest agreement for all devices compared to ECG. Overall, there did not seem to be proportionate bias of consistent increases in error from lower to higher heart rates. Still, systematic and proportionate biases were influenced by exercise type across all devices and should be considered in conjunction with limits of agreement, MAPE, and CCC.

Majority of previous literature has suggested that chest strap devices are superior in validity compared to wrist-worn devices (Cosoli et al., 2020; Etiwy et al., 2019; Gillinov et al., 2017; Pasadyn et al., 2019). In some instances, these chest worn devices are considered the reference device for concurrent validity studies (Chow & Yang, 2020; Düking et al., 2020; Fuller et al., 2020; Hajj-Boutros et al., 2021; Muggeridge et al., 2021). The better validity of chest strap wearables may be due to the modality used to estimate heart rate by sensing electrical activity similar to reference ECG. Chest strap devices may also be more accurate due to less vulnerability to motion artifact (i.e., better maintained position) compared to wrist-worn devices (Murdock & Hagen, 2018). Nonetheless, the H10 has demonstrated strong signal qualities across a variety of exercise intensities and modalities, deeming it a valid device for many movements (Gilgen-Ammann et al., 2019). Some have found chest strap validity to vary across exercise intensities

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with greater underestimations at higher intensities (Boudreaux et al., 2018), while others have noted acceptable validity regardless of exercise intensity or modality (Gillinov et al., 2017). The HRM-Pro and H10 had comparable levels of validity across all protocols, with their lowest, albeit still acceptable, agreement with the reference ECG during Tabata-Circuit. In some circumstances, such as H10 during CYCLE and RUCK-M, the negligible systematic overestimations of the H10 could be accounted for but more research is needed to verify. To ensure the greatest level of validity users should pay close attention to strap wear, battery life, cleaning, and maintenance.

The current results suggest that the Verity Sense may be an accurate alternative to chest straps. The optical PPG armband's agreement with ECG heart rate corresponded with chest strap devices except for Tabata-Circuit, which may result from greater use of excessive arm motions and musculature. Previous literature found the Polar OH1 (prior Verity Sense model) was in strong agreement with the ECG during moderate-to-high intensity treadmill and cycling activities (Hettiarachchi et al., 2019), as well as moderate-to-vigorous Yoga (Schubert et al., 2018). However, other research found the OH1 to accurately estimate heart rate during sedentary, steady cycling, and incremental running to exhaustion, but validity suffered during sprinting tasks on a cycle ergometer and treadmill (Muggeridge et al., 2021). Thus, the Verity should be used with caution during similar training with high emphasis on arm musculature and motions.

Wrist-worn devices in the current study (Fenix-6, Grit-X) were less accurate than chest and armband devices, in agreement with prior findings (Gillinov et al., 2017; Pasadyn et al., 2019). Namely, the Grit-X was considered invalid across all exercises, but the Fenix-6 demonstrated acceptable error rates during RUCK-M and CYCLE. Interestingly, there were several time points in which the Fenix-6 had large overestimations during RUCK-S (Figure 3), but the magnitude of these systematic overestimations was lower (within practically acceptable limits) during RUCK-

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M (Figure 4). This would suggest that the agreement with ECG heart rate was greater for the Fenix-6 during higher-intensity rucking, which may be due to potential for random error. However, the strongest validity for the Fenix-6 occurred during CYCLE where the systematic overestimations were lowest, while the greatest error for the Fenix-6 was noted during Tabata-Circuit.

Within an activity, such as treadmill (Pasadyn et al., 2019), cycle ergometer (Chow & Yang, 2020), and resistance training tasks (Boudreaux et al., 2018; Jo et al., 2016; Müller et al., 2019), it is common for wrist-worn device validity to decrease as exercise intensity increases. However, others have suggested that wrist-worn devices have higher error rates during light-to-moderate treadmill activity intensities (Dooley et al., 2017), which agrees with the present findings. Compared to ECG, consumer wearable devices showed more absolute error during activity than rest with the direction and magnitude of errors dependent on activity type (Bent et al., 2020). Similarly, literature has shown the Garmin Fenix 5 as having higher error rates than the Polar Vantage V when compared to the Polar H7 chest strap during higher-intensity activities (i.e., sprinting) but acceptable error rates at lower-intensity activities such as walking (Düking et al., 2020). Meanwhile, others have found devices, such as Polar Vantage V, to fluctuate in validity across exercises while demonstrating more validity during running and resistance exercises compared to cycling and walking (Hajj-Boutros et al., 2021). Thus, the validity of wearable sensors may also be impacted by the manner in which resistance exercises are performed. As an example, previous research that has found acceptable wrist device validity during resistance exercises carried out controlled tasks (i.e. chest press, leg press, seated cable row) on machines (Hajj-Boutros et al., 2021), while the current Tabata-Circuit included various calisthenics and plyometric-style exercises. The variability found in laboratory settings may bring to question their viability when used in practical outdoor environments with greater movement artifacts.

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Wrist-worn devices typically use PPG, which estimates heart rate from contraction-induced fluctuations in blood volume via light reflections from the device's light-emitting diodes. Coupled with the location of the device, these differences, compared to typical chest straps, are subject to greater sources of error. Error is often noted from motion artifact, which helps explain differences in validity depending on form and movement execution. Although some literature found device validity to be influenced by device fit, ambient light, and skin tone (i.e., darker skin tone showing more susceptibility to error), others have not replicated the effect of skin tone on device validity (Bent et al., 2020; Sañudo et al., 2019). It is also noted in the literature that sex, body mass index, maximal oxygen consumption, skin type, or wrist size did not influence results of validity for optical heart rate sensors (Gillinov et al., 2017; Horton et al., 2017). Thus, it seems the errors may be most affected by activity types and movement artifacts (Bent et al., 2020). Collectively, the Tabata circuit likely lead to the worst outcomes for devices based on motion artefacts from heavy arm movements and greater fluctuations in blood volume from gripping exercises. However, The Tabata circuit also introduced more fluctuations from low to high and high to low heart rates in a rapid manner which could lead to potential second-by-second heart rate disagreements due to any delay in signal recordings. For these reasons, along with previous suggestions (Gillinov et al., 2017; Navalta et al., 2020), we suggest using electrode-based chest straps (or PPG arm bands under the correct circumstances) when accurate continuous measures of heart rate are needed for real-time feedback.

Results of the current study only apply to the heart rate monitors included herein, which were chosen based on use within tactical and athletic populations that train in heart rate zones. Further, devices were selected based on the most recent offerings, at the time, from two popular wearable sensor manufacturers. The comparisons were made in a small sample ($n = 8$); thus, limiting the

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3 358 generalizability to a wider population with interindividual variations that may affect the results.
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5 359 Therefore, these results are only representative of individuals with similar descriptions (i.e., body
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7 360 composition, age) as the current healthy volunteers. Still, the current sample permitted adequate
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9 361 power for comparisons of second-by-second heart rate comparisons (n=4500-8000) which was the
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11 362 intent of the current procedures. These results may not be repeated during different activities or in
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13 363 real world settings, such as sports participation or tactical events, including rucking with more
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15 364 equipment in various terrains outside of the laboratory. Next steps should also include validation
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17 365 of summary metrics (i.e., max/ average heart rate) from the entire session to provide insight on the
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19 366 overall training demands from each session. Finally, due to the high validity and high usability of
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21 367 the Polar H10 as shown in these results, the device may serve as the reference during ecological
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23 368 validity testing environments, such as real-world tactical training, sport training/competition.
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29 369 **PRACTICAL APPLICATIONS**

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32 370 These results are crucial for the utilization of wearables by human performance practitioners
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34 371 and users, as they detail levels of validity of popular manufacturers. Individuals must understand
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36 372 these devices are not always accurate readings such that users are left with their best judgment to
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38 373 understand whether the heart rate readings are accurate at any given time. However, according to
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40 374 these findings, the chest strap and arm-band devices provide the most accurate readings. With the
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42 375 high validity and usability of the Polar H10 chest strap, further research is encouraged to compare
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44 376 devices in real world settings, such as tactical training. It is important to also understand that the
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46 377 chest strap devices may also be recorded to a wrist-watch device rather than a smart phone
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48 378 application and the arm-band (Polar Verity) may be recorded locally to the device and synced to
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50 379 the smart phone application after the exercise. These features improve the feasibility of using these
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54 380 devices during training when smart phone applications are not optimal. It is important for users to
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381 ensure the sampling rates are consistently used as some devices allow sampling rates to be changed
382 from every second to every five seconds for example.

383 Research validating wearable sensors should also track across a range of movements since the
384 validity of devices in the current study were dependent upon activity and intensity (i.e., heart rate).
385 Nonetheless, the H10 was the most accurate device included in the current study. Acceptable
386 agreement with the ECG reference device was also noted for the HRM-Pro with slight
387 underestimations (-1.53 to -1.69 bpm) and the Verity with slight overestimations (1.69 to 1.77
388 bpm) during the rucking tasks. However, the Verity noted increased error during Tabata-Circuit,
389 along with wrist-worn devices (Fenix-6 and Grit-X), which likely resulted from additional
390 movement artifacts. Overall, the wrist-worn devices demonstrated the weakest agreement, but the
391 Fenix-6 outperformed the Grit-X and showed acceptable agreement during RUCK-M and CYCLE.
392 Thus, when accurate real-time heart rate measures are required for training, chest strap devices
393 (i.e., H10 and HRM-Pro) should be used, while armband devices may be appropriate alternatives
394 under certain conditions.

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400 Disclosure statement.

401 The authors report there are no competing interests to declare.

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Figure 1. Descriptive diagram of protocols within each testing session.

Figure 2. Bland-Altman plots for each device compared to electrocardiography during steady-state treadmill rucking. Each black dot represents a second-by-second heart rate comparison where the dark gray, light gray (upper bound), and mid gray (lower bound) shaded areas are the respective value and 95% confidence intervals surrounding the mean, upper, and lower 95% confidence interval, respectively. The solid gray line with shaded gray area represents the slope and the respective bounds.

Figure 3. Bland-Altman plots for each device compared to electrocardiography during maximal incremental treadmill rucking. Each black dot represents a second-by-second heart rate comparison where the dark gray, light gray (upper bound), and mid gray (lower bound) shaded areas are the respective value and 95% confidence intervals surrounding the mean, upper, and lower 95% confidence interval, respectively. The solid gray line with shaded gray area represents the slope and the respective bounds.

Figure 4. Bland-Altman plots for each device compared to electrocardiography during YMCA submaximal cycling. Each black dot represents a second-by-second heart rate comparison where the dark gray, light gray (upper bound), and mid gray (lower bound) shaded areas are the respective value and 95% confidence intervals surrounding the mean, upper, and lower 95% confidence interval, respectively. The solid gray line with shaded gray area represents the slope and the respective bounds.

Figure 5. Bland-Altman plots for each device compared to electrocardiography during high-intensity Tabata Circuit. Each black dot represents a second-by-second heart rate comparison where the dark gray, light gray (upper bound), and mid gray (lower bound) shaded areas are the respective value and 95% confidence intervals surrounding the mean, upper, and lower 95%

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confidence interval, respectively. The solid gray line with shaded gray area represents the slope and the respective bounds.

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Table 1. Comparisons of heart rate monitors to electrocardiography during steady-state treadmill rucking.

Statistic	Polar H10	Garmin HRM-Pro	Polar Verity	Garmin Fenix 6	Polar Grit X
N	7167	6517	7200	5749	5350
OLP R ²	0.983	0.910	0.977	0.612*	0.477*
OLP Intercept	0.358	-1.531	1.690	-1.952	12.70
	(-0.015, 0.731)	(-2.409, -0.654)*	(1.270, 2.110)*	(-4.025, 0.121)	(10.60, 14.90)*
OLP Slope	0.999	0.991	0.991	1.060	0.977
	(0.996, 1.002)	(0.984, 0.999)*	(0.987, 0.994)*	(1.040, 1.080)*	(0.959, 0.997)*
CCC (r _c)	0.991	0.948	0.988	0.759*	0.638*
	(0.991, 0.992)	(0.946, 0.951)	(0.987, 0.988)	(0.748, 0.770)	(0.623, 0.653)
MAPE (%)	1.95±2.36 ^{b,d,e}	3.73±6.32 ^{a,c,d,e}	2.20±2.68 ^{b,d,e}	5.44±8.82 ^{a,b,c,e}	9.26±13.32 ^{a,b,c,d}

N; number of second-by-second heart rate comparisons; OLP R²; Ordinary Least Products Regression (OLP) coefficient; OLP Intercept; systematic bias was present if 95% confidence interval did not include “0”; OLP Slope; proportional bias was present if 95% confidence interval did not include “1.0”. CCC, Lin’s Concordance Coefficients with the following cut offs < 0.80: unacceptable, 0.80-0.89: poor, 0.90 to 0.94: moderate, 0.95 to 0.99: substantial, > 0.99 almost perfect; MAPE; mean absolute percent error (95% confidence interval) > 5% was considered unacceptable. *, indicates below acceptable threshold for the respective statistical test.

^a, significantly different than Polar H10; ^b, significantly different than Garmin HRM Pro; ^c, significantly different than Polar Verity; ^d, significantly different than Garmin Fenix 6; ^e, significantly different than Polar Grit X

Table 2. Comparisons of heart rate monitors to electrocardiography during incremental maximal treadmill rucking.

Statistic	Polar H10	Garmin HRM-Pro	Polar Verity	Garmin Fenix 6	Polar Grit X
N	7163	6675	7410	4977	6164
OLP R ²	0.995	0.988	0.991	0.926	0.453
OLP Intercept	0.375	-2.63	1.77	6.82	38.10
	(0.127, 0.623)*	(-3.01, -2.24)*	(1.45, 2.08)*	(5.78, 7.87)*	(36.10, 40.10)*
OLP Slope	0.999	1.010	0.996	0.980	0.867
	(0.997, 1.001)	(1.008, 1.012)*	(0.994, 0.998)*	(0.973, 0.988)*	(0.851, 0.883)*
CCC (r _c)	0.997	0.993	0.995	0.954	0.553*
	(0.997, 0.997)	(0.993, 0.994)	(0.995, 0.995)	(0.952, 0.957)	(0.538, 0.567)
MAPE (%)	1.28±1.90 ^{b,c,d,e}	1.96±3.60 ^{a,d,e}	1.84±2.25 ^{a,d,e}	4.44±5.82 ^{a,b,c,e}	16.45±17.59 ^{a,b,c,d}

N; number of second-by-second heart rate comparisons; OLP R²; Ordinary Least Products Regression (OLP) coefficient; OLP Intercept; systematic bias was present if 95% confidence interval did not include “0”; OLP Slope; proportional bias was present if 95% confidence interval did not include “1.0”. CCC, Lin’s Concordance Coefficients with the following cut offs < 0.80: unacceptable, 0.80-0.89: poor, 0.90 to 0.94: moderate, 0.95 to 0.99: substantial, > 0.99 almost perfect; MAPE; mean absolute percent error (95% confidence interval) > 5% was considered unacceptable. *, indicates below acceptable threshold for the respective statistical test.

^a, significantly different than Polar H10; ^b, significantly different than Garmin HRM Pro; ^c, significantly different than Polar Verity; ^d, significantly different than Garmin Fenix 6; ^e, significantly different than Polar Grit X

Table 3. Comparisons of heart rate monitors to electrocardiography during YMCA submaximal cycling.

Statistic	Polar H10	Garmin HRM-Pro	Polar Verity	Garmin Fenix 6	Polar Grit X
N	7158	5866	7128	6464	7567
OLP R ²	0.974	0.966	0.968	0.930	0.685*
OLP Intercept	0.498	0.055	0.145	1.430	-0.228
	(0.047,0.950)*	(-0.503, 0.612)	(-0.355, 0.644)	(0.673, 2.187)*	(-1.730, 1.275)
OLP Slope	0.999	0.994	1.01	0.990	1.08
	(0.996, 1.003)	(0.990, 0.999)*	(1.00, 1.01)*	(0.984, 0.997)*	(1.07, 1.10)*
CCC (r _c)	0.987	0.983	0.983	0.964	0.777*
	(0.986, 0.987)	0.982, 0.983)	(0.982, 0.984)	(0.962, 0.966)	(0.768, 0.785)
MAPE (%)	2.67±3.61 ^{c,d,e}	2.85±4.06 ^{c,d,e}	3.16±3.91 ^{a,b,d,e}	4.23±5.37 ^{a,b,c,e}	8.49±9.92 ^{a,b,c,d}

N; number of second-by-second heart rate comparisons: OLP R²; Ordinary Least Products Regression (OLP) coefficient: OLP Intercept; systematic bias was present if 95% confidence interval did not include “0”: OLP Slope; proportional bias was present if 95% confidence interval did not include “1.0”. CCC, Lin’s Concordance Coefficients with the following cut offs < 0.80: unacceptable, 0.80-0.89: poor, 0.90 to 0.94: moderate, 0.95 to 0.99: substantial, > 0.99 almost perfect: MAPE; mean absolute percent error (95% confidence interval) > 5% was considered unacceptable. *, indicates below acceptable threshold for the respective statistical test.

^a, significantly different than Polar H10; ^b, significantly different than Garmin HRM Pro; ^c, significantly different than Polar Verity; ^d, significantly different than Garmin Fenix 6; ^e, significantly different than Polar Grit X

Table 4. Comparisons of heart rate monitors to electrocardiography during Tabata Style Circuit Training.

Statistic	Polar H10	Garmin HRM-Pro	Polar Verity	Garmin Fenix 6	Polar Grit X
N	6658	6680	5751	4618	6876
OLP R ²	0.862	0.829	0.708	0.390*	0.103*
OLP Intercept	1.06	-4.05	14.25	14.73	28.7
	(-0.27, 2.40)	(-5.57, -2.53)*	(12.30, 16.20)*	(11.63, 17.84)	(26.0, 31.4)
OLP Slope	0.990	1.02	0.948	0.948	0.906
	(0.981, 0.999)*	(1.01, 1.03)*	(0.927, 0.969)*	(0.927, 0.969)*	(0.886, 0.926)*
CCC (r _c)	0.928	0.909	0.836	0.589*	0.243*
	(0.925, 0.932)	(0.905, 0.913)	(0.828, 0.844)	(0.571, 0.606)	(0.226, 0.26)
MAPE (%)	3.40±5.09 ^{c,d,e}	3.35±5.44 ^{c,d,e}	5.36±6.37 ^{a,b,d,e}	10.21±9.11 ^{a,b,c,e}	14.16±12.05 ^{a,b,c,d}

N; number of second-by-second heart rate comparisons: OLP R²; Ordinary Least Products Regression (OLP) coefficient: OLP Intercept; systematic bias was present if 95% confidence interval did not include “0”: OLP Slope; proportional bias was present if 95% confidence interval did not include “1.0”. CCC, Lin’s Concordance Coefficients with the following cut offs < 0.80: unacceptable, 0.80-0.89: poor, 0.90 to 0.94: moderate, 0.95 to 0.99: substantial, > 0.99 almost perfect: MAPE; mean absolute percent error (95% confidence interval) > 5% was considered unacceptable. *, indicates below acceptable threshold for the respective statistical test.

^a, significantly different than Polar H10; ^b, significantly different than Garmin HRM Pro; ^c, significantly different than Polar Verity; ^d, significantly different than Garmin Fenix 6; ^e, significantly different than Polar Grit X

Two separate R (Rucking) Sessions			Two separate C (Cycle and Circuit) Sessions			
Protocol 1: Treadmill Rucking at Steady State			Cycling 50 rpm for 3-minute stages with additional 4 th minute if heart rate from minutes 2 to 3 is > 5 bpm difference.			
Time (min)	Speed (kph)	Grade (%)	Warm-up	0 kg		
0	4.8	0	1 st Stage	0.5 kg (the final HR determines resistance below)		
3	6.4	0				
6	6.4	0		HR < 80 bpm	HR 80-89 bpm	HR 90-100 bpm
9	6.4	0				
12	6.4	0				
<i>(rest ~10 min)</i> Protocol 2: Treadmill Rucking to Voluntary Exhaustion						
Time (min)	Speed (kph)	Grade (%)	2 nd Stage	2.5 kg	2.0 kg	1.5 kg
			3 rd Stage	3.0 kg	2.5 kg	2.0 kg
			4 th Stage	3.5 kg	3.0 kg	2.5 kg
0	4.8	0	<i>(rest ~10 min)</i> Tabata Circuit			
3	4.8	5	3 complete rounds. All movements performed for 20 seconds followed by 10 seconds of rest.			
6	6.4	5	Group A: Pull-ups, Kettlebell Swings, Push-ups, Farmer's Carry			
9	6.4	10	Rest for 1 minute			
12	6.4	12.5	Group B: Burpees, sit-ups, Box Jumps (24 inches), Air Squats			
15	6.4	15	Rest for 1 minute			
18	7.2	15				
21	7.2	15				

Figure 1

329x357mm (47 x 47 DPI)

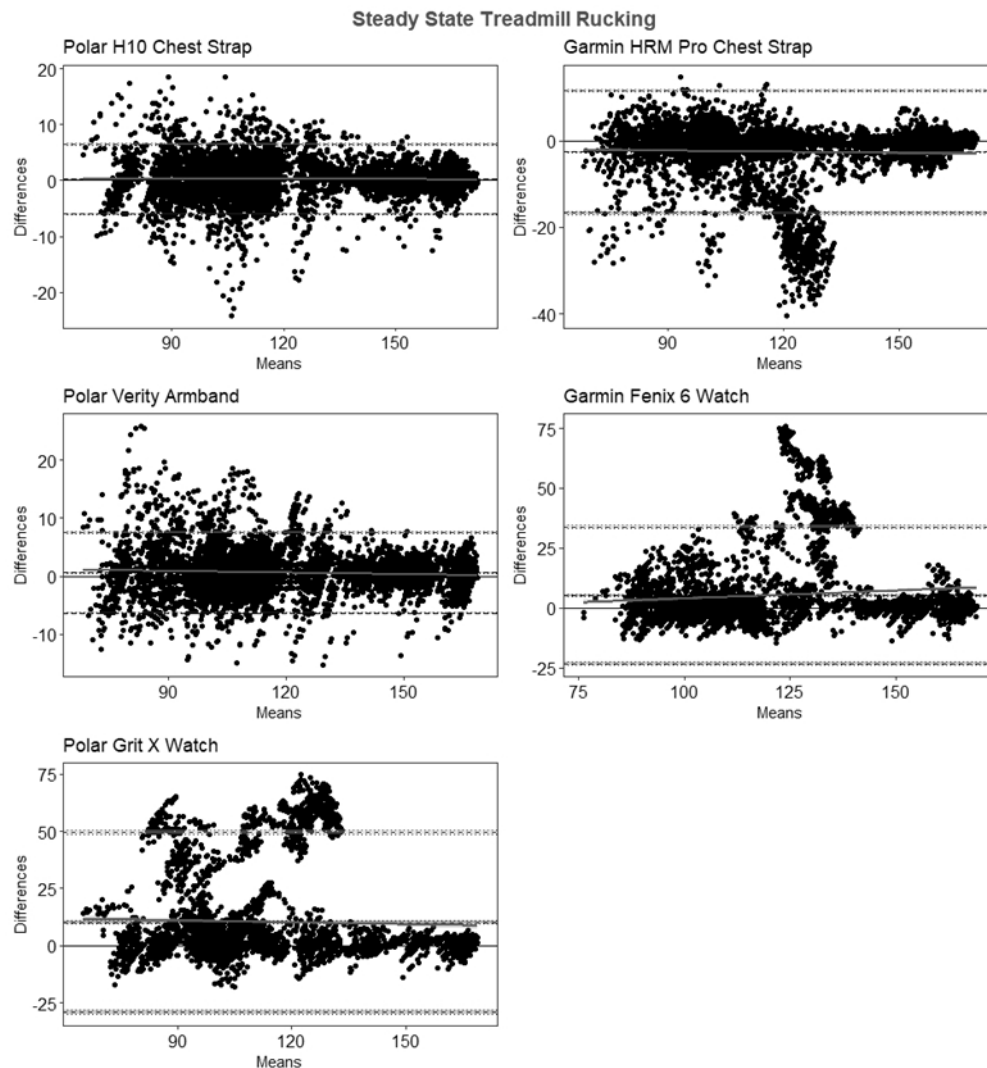


Figure 2

421x452mm (47 x 47 DPI)

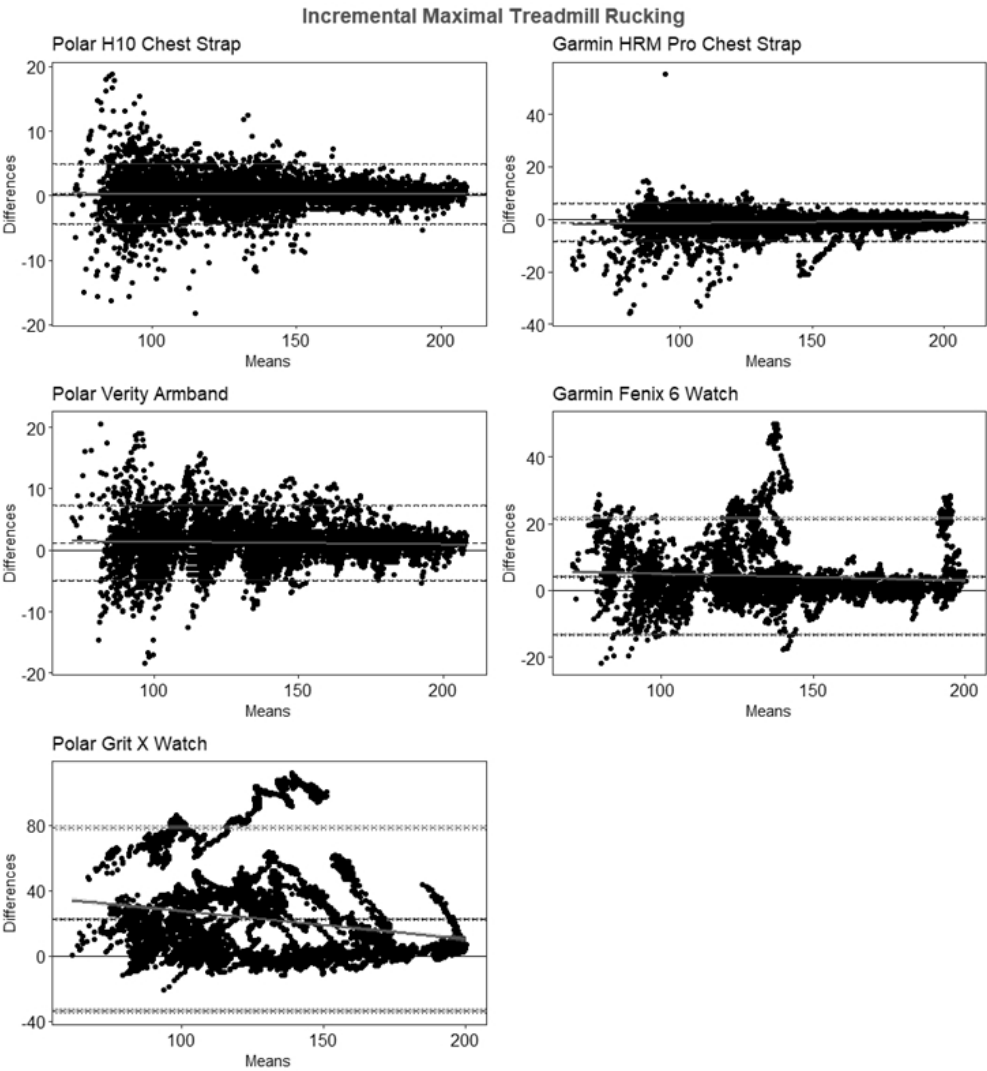


Figure 3

396x425mm (47 x 47 DPI)

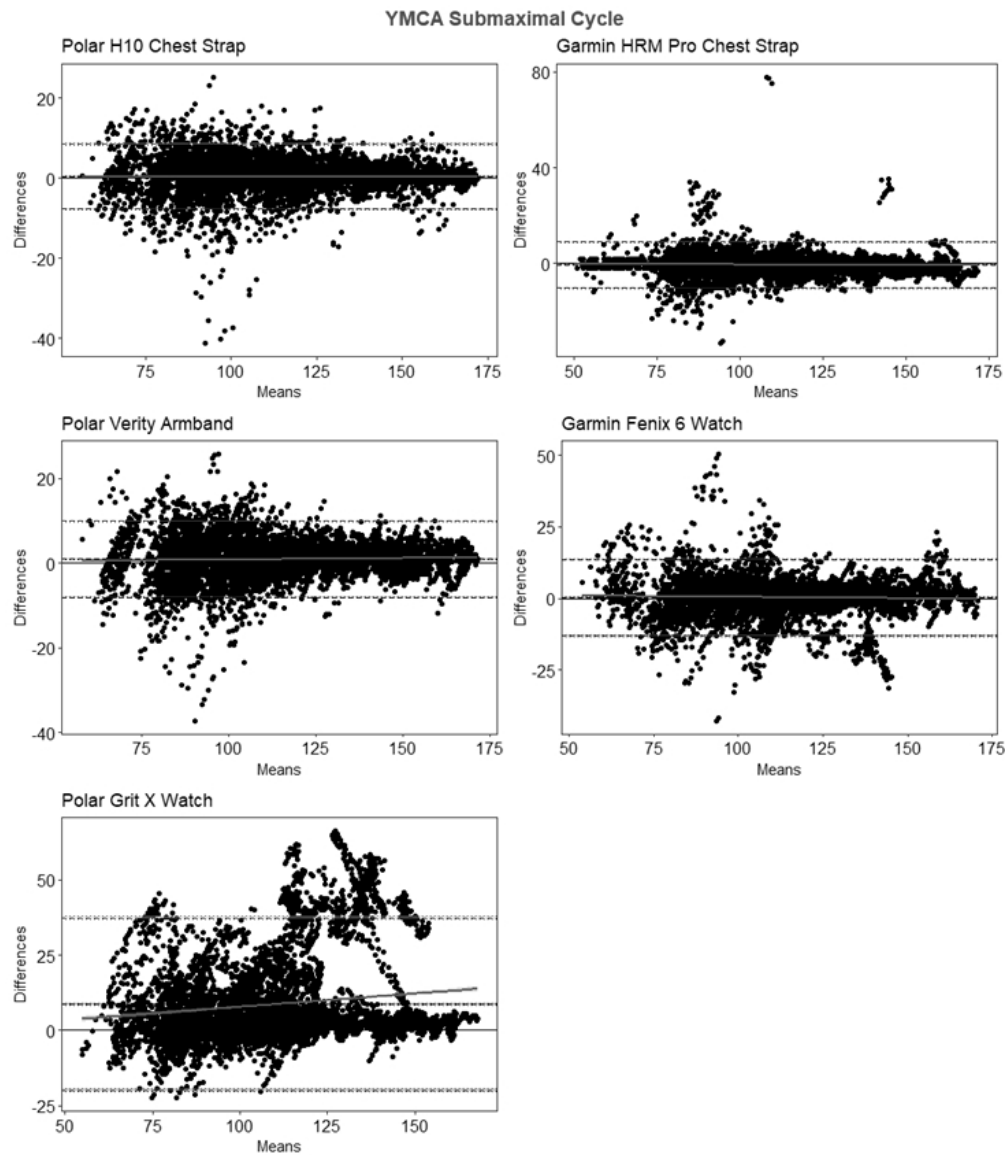


Figure 4

395x456mm (47 x 47 DPI)

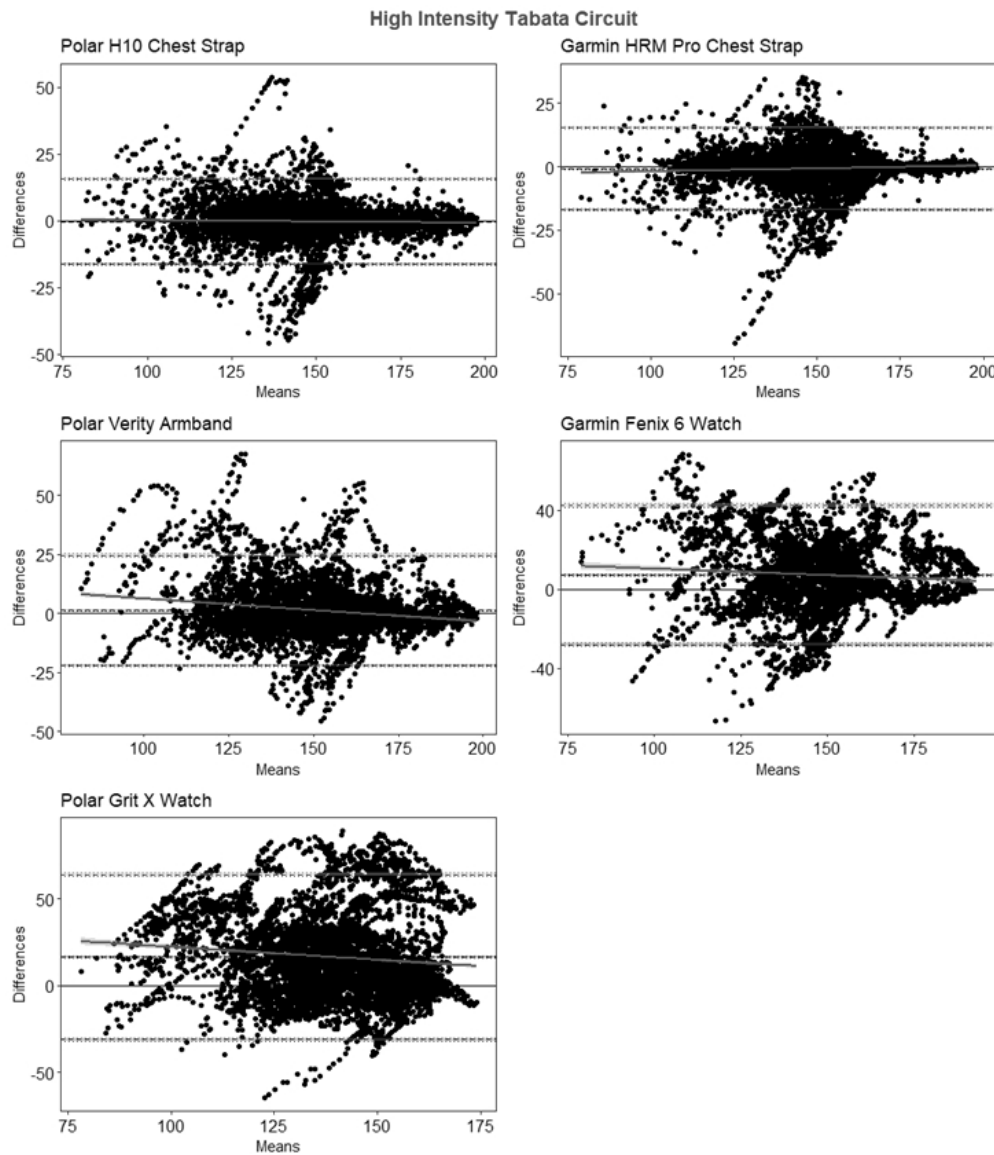


Figure 5
393x454mm (47 x 47 DPI)